HWA CHONG INSTITUTION

JC2 PRELIMINARY EXAMINATIONS

H3 BIOLOGY 9815

17 Sept 2008 **2 hr 30 min**

INSTRUCTIONS TO CANDIDATES

Check that there are **16** printed pages in this question booklet, excluding the cover page.

Write your answers on the answer provided.

Begin each question on a fresh sheet of paper.

SECTION A

This section contains five questions. Answer **ALL** the questions.

SECTION B

This section contains four questions. Answer **three** out of the **four** questions.

SECTION C

This section contains one question. Answer the question.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [ ] at the at the end of each question.

A table of amino acids is provided on **page 17**.

Section A

Answer all questions in this section.

QUESTION 1

Table 1.1 shows the different types of mutant hemoglobin protein resulting from various amino acid substitutions in the β chain of human hemoglobin. The net change in average charge of each mutant hemoglobin at physiological pH is also shown.

Mutant hemoglobin	Substitution position in β chain	Normal hemoglobin	Mutant hemoglobin	Net change in average charge
		Residue	Replacement residue	
C	6	Glu	Lys	
$G_{San\ Jose}$	7	Glu	Gly	+1
$M_{Milwaukee}$	67	Val	Glu	
Zurich	63	His	Arg	0

Table 1.1

(a) State the net change in average charge for mutant hemoglobin C and $M_{Milwaukee}$. [1]

(b) A sample of the hemoglobin of a patient known to be suffering from a genetic defect in the β chain is subjected to electrophoresis at pH 8 along with a sample of normal hemoglobin A. The patient's hemoglobin migrates towards the anode faster than hemoglobin A.

Which of the β-chain mutants listed in **Table 1.1** could the patient's hemoglobin be? Briefly explain your answer. [2]

(c) Given that position 67 lies in the cleft of the β chain where the heme functional group is wedged, deduce if the substitution of Glu for Val in $M_{Milwaukee}$ leads to any effect on the stability of the hemoglobin molecule. Justify your answer. [2]

(d) The relative molecular mass of human hemoglobin is 68000 and the molecules are contained in red blood cells. The relative molecular mass of the hemoglobin of a species of worm is 2.85 million and the molecules are found in the plasma. Suggest one advantage of the relatively large size of hemoglobin when it is in the plasma. [1]

(e) Hemoglobin is an intracellular protein consisting of α and β subunits predominantly made up of lengths of helical regions. Give an example of an extracellular protein that has a helical conformation, and discuss the significance of the extracellular protein's helical nature to its function. [4]

[Total: 10 marks]

QUESTION 2

Student **X** is developing a purification scheme for an enzyme. Her purification steps are shown in **Table 2.1**. The first step is a 40% saturated $(NH_4)_2SO_4$ precipitation. The precipitated protein is subjected to anion exchange chromatography (DEAE). The starting buffer is at pH 7.0 and the protein does not bind to the column. The flow-through fraction is then subjected to preparative isoelectric focusing using a pH range of 8.0 - 10.0. The fraction at pH 9.7 contains the enzyme.

Students **Y** and **Z** also attempted to purify the enzyme but modified Student **X**'s original purification protocol.

(a) Calculate the specific activity, fold-purification and the percentage yield for each step as compared to the initial starting material and fill in the values in **Table 2.1** on the insert provided. [3]

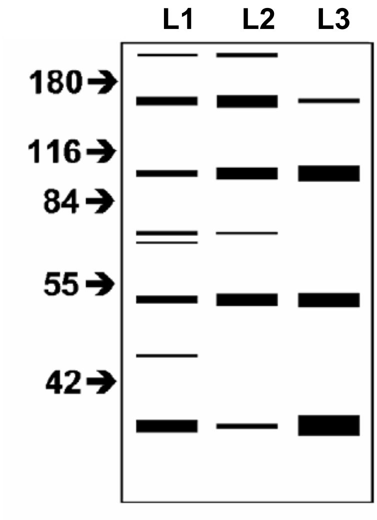
Steps	**TOTAL RECOVERY**		**SPECIFIC ACTIVITY**	**FOLD PURIFIED**	**% YIELD**
	PROTEIN (mg)	**ENZYME (units)**			
Student X					
Initial starting material	511	4200	8.2	-	-
40 % $(NH_4)_2SO_4$	181	3991			
DEAE flow-through, pH 7.0	67.6	3787			
Isoelectric focusing (pH 9.7)	3.8	2949			
Student Y					
Initial starting material	714	5880	8.2	-	-
40 % $(NH_4)_2SO_4$	253	5584			
DEAE flow-through, pH 9.0	5.1	4800			
Student Z					
Initial starting material	410	3360	8.2	-	-
40 % $(NH_4)_2SO_4$	146	3192			
Cation exchange chromatography, pH 7.0 NaCl elution	1.9	2912			

Table 2.1

(b) State which student has the best purification scheme and give a reason for your answer. [2]

(c) Based on Student **X**'s purification scheme, account for Student **Z**'s decision to use cation exchange chromatography. [1]

Student **Z** decided to test the reliability of her purification scheme with SDS-PAGE. The results are shown in **Fig. 2.1**.



Key:

Lane 1 (L1) - Initial starting material

Lane 2 (L2) - After 40 % $(NH_4)_2SO_4$

Lane 3 (L3) - After CM, pH 7.0 NaCl elution

Fig. 2.1

(d) State the approximate size(s) of the protein(s) which is/are likely to be responsible for the enzyme activity. Give a reason for your answer. [2]

(e) Describe the technique which would enable Student **X** to elucidate real-time interaction of the enzyme with other proteins and the kinetics of these interactions. [2]

[Total: 10 marks]

QUESTION 3

In an experiment on human mesenchymal stem cells (hMSCs), hMSC population **1** was grown in the presence of tumour growth factor β (TGF-β) while population **2** was not. Their total proteins were extracted for further analysis as partly shown in **Fig. 3.1**.

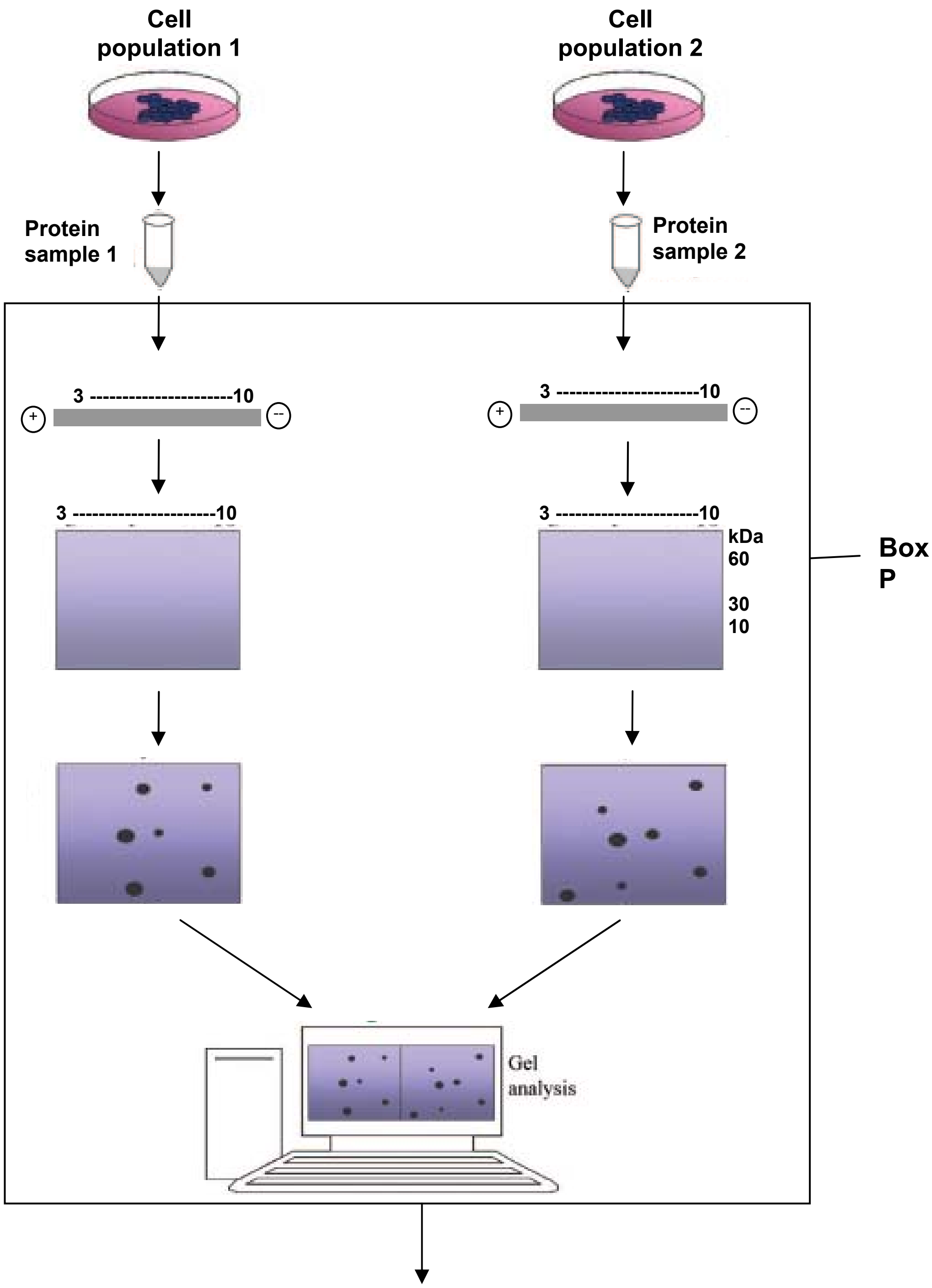


Fig. 3.1

(a) Briefly describe the steps highlighted in Box **P**. [2]

(b) Outline the subsequent steps not shown in **Fig. 3.1**, and explain how the results of this experiment may help researchers understand the effects of TGF-β on hMSCs. [4]

(c) State and explain a drawback of the experimental methodology shown in **Fig. 3.1**. [2]

(d) Briefly describe another methodology that allows the simultaneous analysis of thousands of proteins. [2]

[Total: 10 marks]

QUESTION 4

Sphingosine-1-phosphate (SPP) is a lipid that is important for cell survival. The synthesis of SPP from sphingosine and ATP is catalysed by the enzyme sphingosine kinase. The velocity of the sphingosine kinase reaction was measured in the presence and absence of 10μM *threo*-sphingosine, an inhibitor that is structurally similar to sphingosine. The results are shown in **Table 4.1**.

[Sphingosine] / μM	**v_0 / mgmin^{-1} (without *threo*-sphingosine)**	**v_0 / mgmin^{-1} (with *threo*-sphingosine)**
2.5	32.3	8.5
3.5	40.0	11.5
5.0	50.8	14.6
10.0	72.0	25.4
20.0	87.7	43.9
50.0	115.4	70.8

Table 4.1

(a) (i) Using the Lineweaver-Burk plot, determine the apparent k_m and V_{max} values in the presence and absence of *threo*-sphingosine. Present your answer on the graph paper provided. [3]

(ii) Account for the difference in V_{max} values in the presence and absence of *threo*-sphingosine. [3]

(b) Using the data provided, estimate k_I, showing all relevant working. [2]

(c) Using the values obtained in **(a) (i)** and **(b)**, comment on the effectiveness of *threo*-sphingosine as an inhibitor of the sphingosine kinase reactions. [2]

[Total: 10 marks]

QUESTION 5

An acetylcholine receptor (abbreviated AChR) is an integral membrane protein that responds to the binding of the neurotransmitter acetylcholine. It is important for the functioning of the nervous system. The structure of acetylcholine is shown in **Fig. 5.1**.

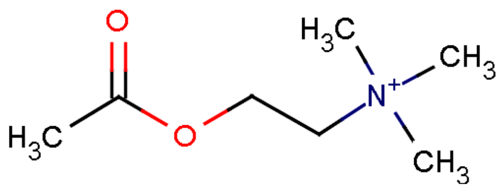


Fig. 5.1

(a) Briefly explain why acetylcholine receptors are necessarily transmembrane proteins. [2]

Although all acetylcholine receptors, by definition, respond to acetylcholine, they respond to other molecules as well. Two classes of acetylcholine receptors exist:

- nicotinic acetylcholine receptors (nAChR) are particularly responsive to nicotine
- muscarinic acetylcholine receptors (mAChR) are particularly responsive to muscarine.

The nAChRs are ligand-gated ion channel receptors, while the mAChRs are G-protein coupled receptors. The two receptors have very little DNA and amino acid sequence homology.

(b) (i) Account for the fact that both nAChRs and mAChRs can bind to acetylcholine even though their primary sequences differ greatly. [2]

(ii) State one structural feature of an mAChR that is not expected to be found in an nAChR. [1]

nAChRs consist of five differing polypeptide subunits surrounding a central pore in the conformation $\alpha_2\beta\delta\gamma$. Each subunit is made up of four transmembrane α-helices named **M1**-**M4**. The structure of an nAChR is shown in **Fig. 5.2**.

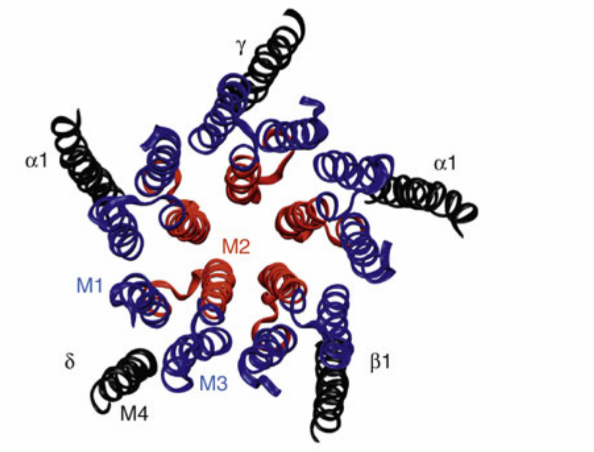


Fig. 5.2

(c) (i) With reference to **Fig 5.2**, explain why there is a large predominance of hydrophobic amino acids in the sequences of helices **M1** and **M3**. [2]

(ii) For all five subunits, helix **M2** is positioned such that it faces the channel pore. Describe and explain one difference that you would expect to see between the amino acid sequences of helices **M2** and **M3**. [2]

The "gate" region of the ligand-gated ion channel consists of five leucine residues that project into the core of the channel, as represented by the black circles in **Fig. 5.3**.

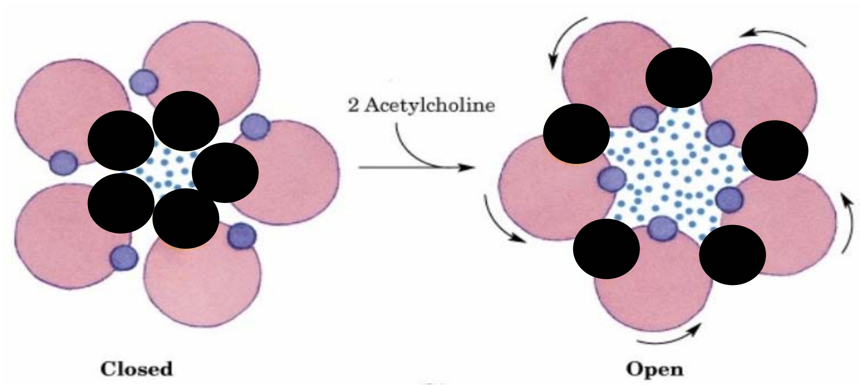


Fig. 5.3

(d) Suggest one reason why leucine is a suitable amino acid for use in the "gate". [1]

[Total: 10 marks]

-- End of Section A --

Section B

Answer three out of the four questions in this section.

QUESTION 6

Fig. 6.1 is a schematic drawing of the structure of an antibody (immunoglobulin) molecule, showing the intramolecular and intermolecular disulphide bridges.

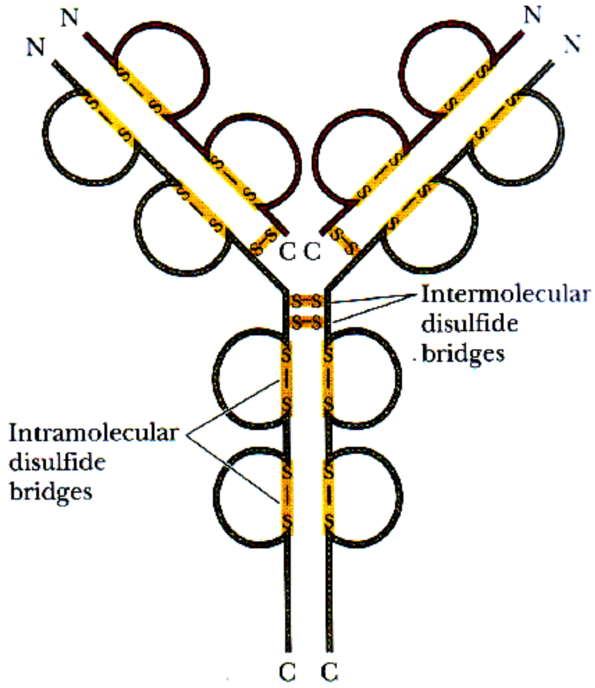


Fig. 6.1

(a) (i) Explain why the immunoglobulin molecule is referred to as a heterotetramer. [2]

(ii) Outline the importance of the disulphide bridges to the structure of the immunoglobulin molecule. [1]

Antibodies possess binding sites that are complementary to specific structural features of antigen epitopes. Each antigen binding site is formed by the N-terminal variable domain of one heavy chain and that of the light chain linked to it, and thus involves amino acid residues from the two variable domains.

Fig. 6.2 is a schematic representation of the complementarity between an antigen epitope and an antigen binding site of an antigen-antibody complex. Numbered letters refer to three different regions of the heavy (H) and light (L) chains, while circled numbers denote the positions of the amino acid residues in the corresponding chain.

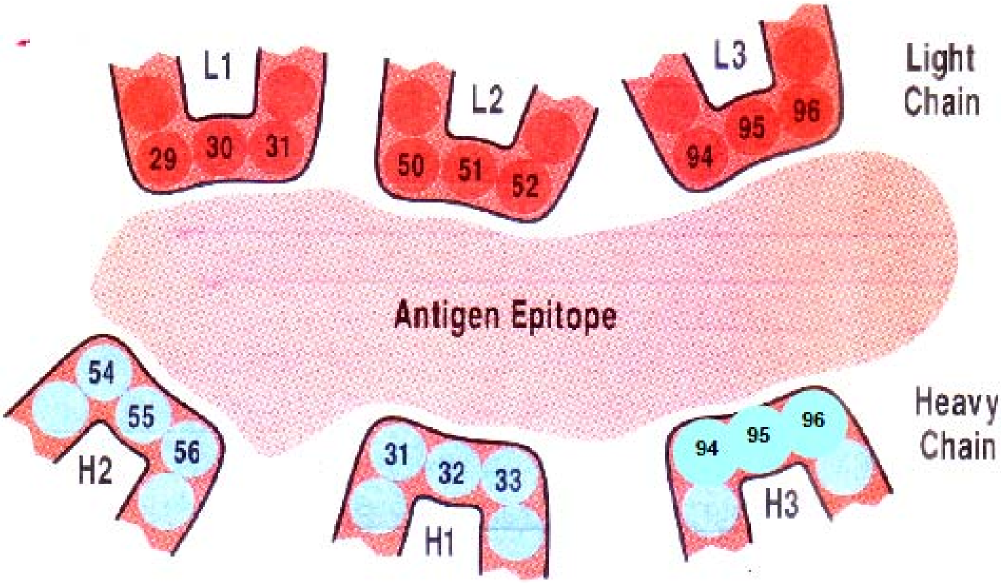


Fig. 6.2

(b) Given that the epitope is protein in nature, describe the bonding interactions that allow for the formation of the antigen-antibody complex shown in **Fig. 6.2**. [2]

The sequences of a large number of immunoglobulin chains are compared, and the variability at each amino acid position is computed as the number of different amino acids found at the particular position, divided by the frequency of the most common amino acid at that position.
The variability of the N-terminal 100 amino acid residues is plotted as shown in **Fig. 6.3**.

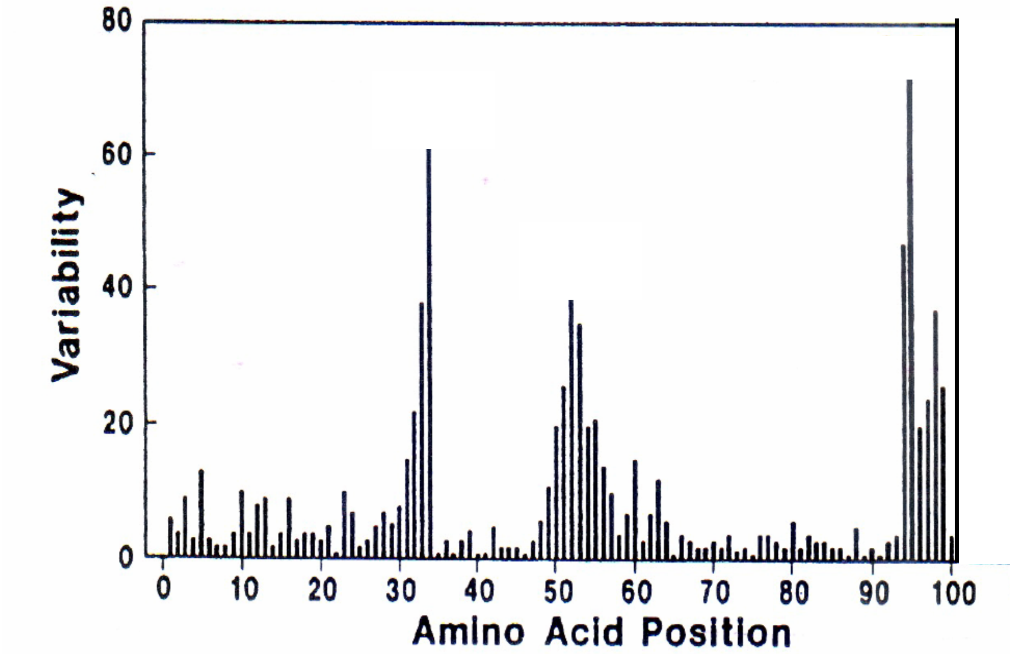


Fig. 6.3

(c) With reference to **Fig. 6.2** and **Fig. 6.3**, relate the relative variability of the amino acid residues at the antigen binding site to their significance in antibody function. [3]

The structure of a class of human immunoglobulins known as IgM is shown in **Fig. 6.4**. The IgM molecule consists of five of the immunoglobulin unit in **Fig. 6.1** joined together by additional disulphide bridges between the F_c portions and by a polypeptide chain called the J chain.

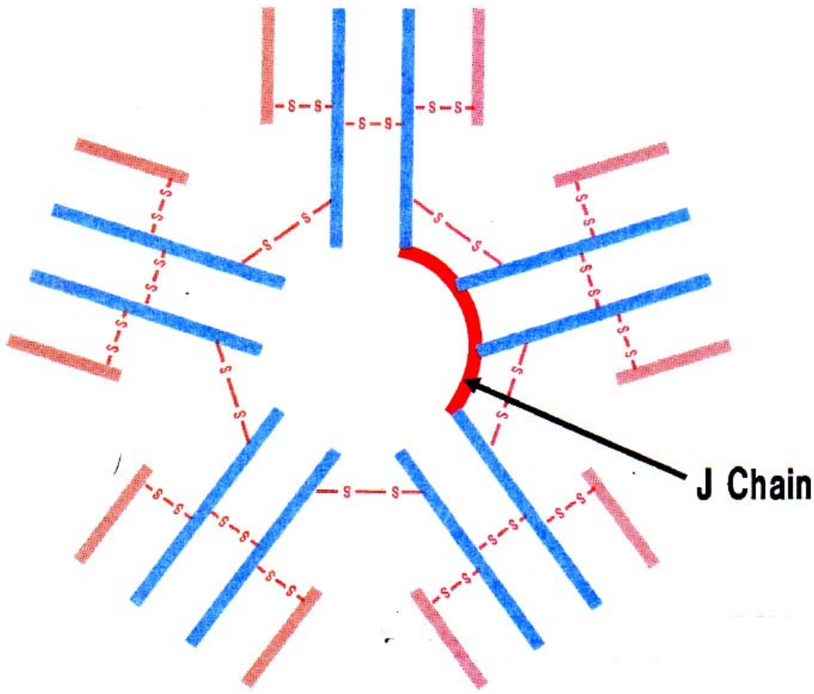


Fig. 6.4

(d) (i) Determine if the IgM molecule is a closed or open structure, and if the interactions between the five units are isologous or heterologous. [1]

(ii) Suggest why IgM is a more efficient antibody than the immunoglobin molecule shown in **Fig. 6.1**. [1]

[Total: 10 marks]

QUESTION 7

To investigate the role of *Yes*, a member of the Src family of kinases, in the cell signalling pathways of early development, a research group compared the phosphoproteomes of wild type donkeyfish embryos with those of *Yes* mutant embryos.

Thirty wild-type and thirty *Yes* mutant embryos at the 24 hours post-fertilisation (24 hpf) stage were lysed. Protein mixtures were digested with trypsin and subsequently labelled at their primary amino groups with normal formaldehyde (CH_2O), "light", for the wild-type embryos or deutero-formaldehyde (CD_2O), "heavy", for the mutant embryos. Samples were mixed in a 1:1 ratio (total peptide content). The resulting peptide mixture was first separated and fractionated by strong cation exchange (SCX) high performance liquid chromatography which, under acidic conditions, causes phosphopeptides to elute earlier than regular tryptic peptides. Resulting SCX fractions were subsequently subjected to MS-MS. These steps are shown in **Fig. 7.1**.

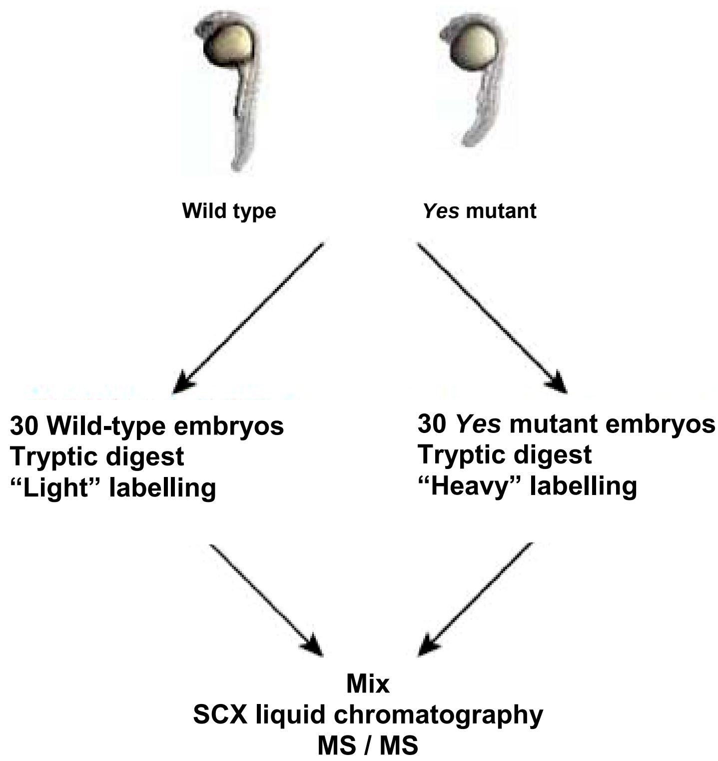


Fig. 7.1

(a) Explain the importance of mixing the samples in a 1:1 ratio after labelling, prior to subsequent analysis. [2]

(b) Describe how the results of this investigation could provide useful information about the role of *Yes* kinase in the developmental events at the 24 hpf stage. [3]

Another research group has decided to employ the RNA interference (RNAi) method to investigate the effects on the donkeyfish embryos if *Yes* kinase is not expressed.

(c) Briefly describe how RNAi is used to suppress the expression of *Yes* kinase protein. [2]

Fig. 7.2 shows the effect of the length of dsRNA on gene expression.

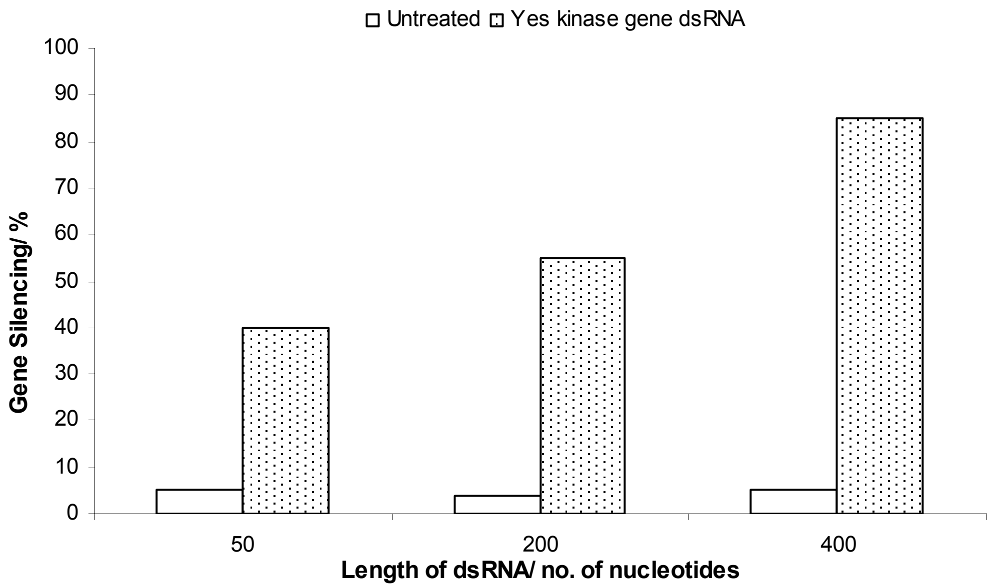


Fig. 7.2

(d) With reference to **Fig. 7.2**, comment on the effectiveness of the length of the dsRNA on the expression of *Yes* kinase. [2]

(e) Suggest why the use of RNAi makes it a valuable research tool in both cell culture and the study of living organisms. [1]

[Total: 10 marks]

QUESTION 8

Enzymes **X** and **Y** catalyse the same reaction and exhibit the rate of reaction versus substrate concentration curves shown in **Fig. 8.1**.

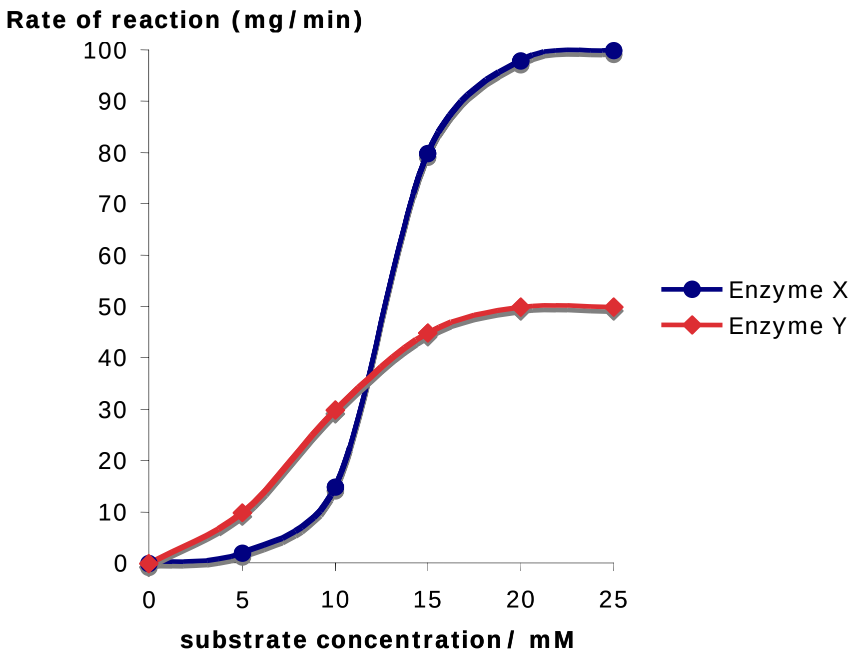


Fig. 8.1

(a) Describe the term "cooperativity". [2]

With reference to **Fig. 8.1**,

(b) (i) Describe the curve exhibited by enzyme **X**. [2]

(ii) Explain the differences in the graphs obtained for enzyme **X** and enzyme **Y**. [4]

(c) Suggest two other factors that can increase the rate of reaction catalysed by enzyme **Y**. [2]

[Total: 10 marks]

QUESTION 9

(a) Describe how ATP synthase couples proton flux to ATP synthesis. [3]

(b) Compounds known as uncouplers are lipid-soluble weak acids that are able to dissolve in the lipid bilayer and create proton channels. Some examples of uncouplers are dinitrophenol (DNP) and valinomycin.

Explain how the addition of an uncoupler to a suspension of mitochondria would affect the action of ATP synthase. [3]

(c) When a proton gradient is generated by coupled electron transfer or ATP hydrolysis, mitochondria take up Ca^{2+} from the external medium. The Ca^{2+} uptake is catalysed by a Ca^{2+} carrier in the inner membrane which exchanges Ca^{2+} for H^+.

Describe the main structural features of this Ca^{2+} carrier that would be necessary for it to carry out its function. [4]

[Total: 10 marks]

-- End of Section B --

Section C

Answer the question.

QUESTION 10

Discuss, with named examples, how the following interactions are essential to cellular function.

(i) protein-protein interactions [8]

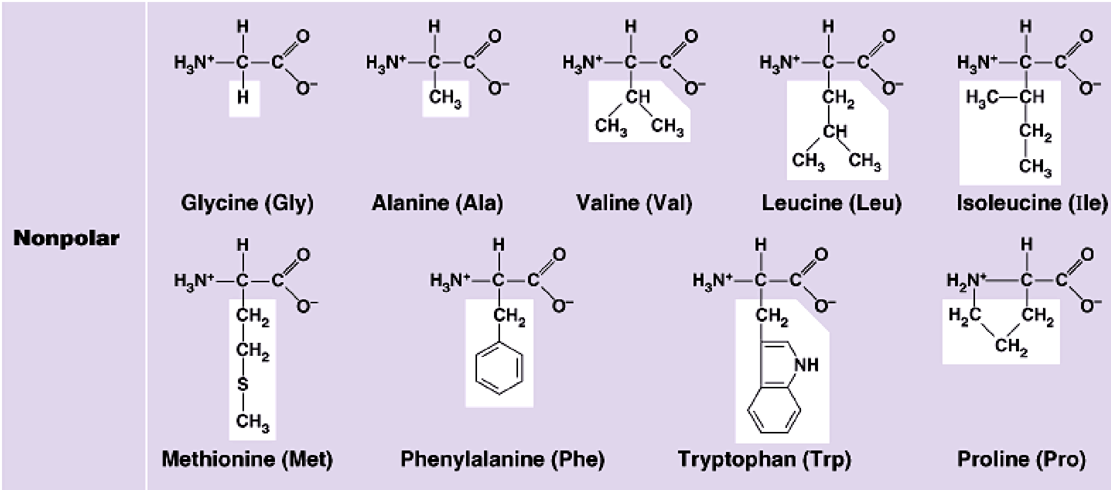
(ii) protein-nucleic acid interactions [12]

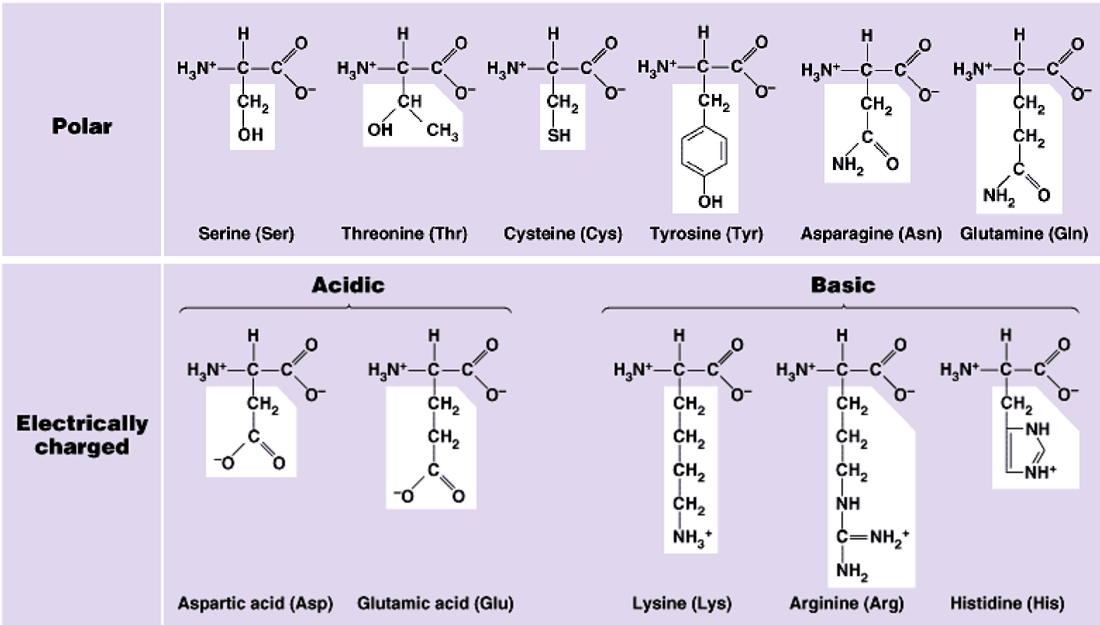
[Total: 20 marks]

-- End of Section C --

-- End of Paper --

Appendix 1: Table of amino acids





Twenty standard Amino Acids